

Earley Parsing and Probabilistic Tree Evaluation

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1 Task 1: Earley Parsing of “time flies like an arrow”

1.1 Sentence and Grammar

The target sentence for parsing is: “time flies like an arrow”

The following grammar (ignoring probabilities for this task) was utilized:

$$\begin{aligned} S &\rightarrow NP VP \\ NP &\rightarrow N \mid Det N \mid NP PP \\ VP &\rightarrow V NP \mid V PP \\ PP &\rightarrow P NP \\ Det &\rightarrow \text{an} \\ N &\rightarrow \text{time} \mid \text{flies} \mid \text{arrow} \\ V &\rightarrow \text{time} \mid \text{flies} \mid \text{like} \\ P &\rightarrow \text{like} \end{aligned}$$

1.2 Final Earley Chart

Chart[0]

- $\gamma \rightarrow \bullet S$ [0]
- $S \rightarrow \bullet NP VP$ [0]
- $NP \rightarrow \bullet N$ [0]
- $NP \rightarrow \bullet Det N$ [0]
- $N \rightarrow \bullet \text{time}$ [0]
- $Det \rightarrow \bullet \text{an}$ [0]

Chart[1] (after scanning “time”)

- $N \rightarrow \text{time} \bullet$ [0]
- $NP \rightarrow N \bullet$ [0]
- $S \rightarrow NP \bullet VP$ [0]
- $VP \rightarrow \bullet V NP$ [1]
- $VP \rightarrow \bullet V PP$ [1]
- $V \rightarrow \bullet \text{flies}$ [1]

*Code available at: <https://github.com/argamas23/Computational-Psycholinguistics-Assignment-5>

- $V \rightarrow \bullet \text{like}$ [1]
- $V \rightarrow \bullet \text{time}$ [1]

Chart[2] (after scanning “flies”)

- $V \rightarrow \text{flies} \bullet$ [1]
- $VP \rightarrow V \bullet NP$ [1]
- $VP \rightarrow V \bullet PP$ [1]
- $NP \rightarrow \bullet N$ [2]
- $NP \rightarrow \bullet \text{Det } N$ [2]
- $N \rightarrow \bullet \text{flies}$ [2]
- $\text{Det} \rightarrow \bullet \text{an}$ [2]
- $PP \rightarrow \bullet P NP$ [2]
- $P \rightarrow \bullet \text{like}$ [2]

Chart[3] (after scanning “like”)

- $P \rightarrow \text{like} \bullet$ [2]
- $PP \rightarrow P \bullet NP$ [2]
- $NP \rightarrow \bullet N$ [3]
- $NP \rightarrow \bullet \text{Det } N$ [3]
- $NP \rightarrow \bullet NP PP$ [3]
- $N \rightarrow \bullet \text{arrow}$ [3]
- $\text{Det} \rightarrow \bullet \text{an}$ [3]

Chart[4] (after scanning “an”)

- $\text{Det} \rightarrow \text{an} \bullet$ [3]
- $NP \rightarrow \text{Det} \bullet N$ [3]
- $N \rightarrow \bullet \text{arrow}$ [4]

Chart[5] (after scanning “arrow”)

- $N \rightarrow \text{arrow} \bullet$ [4]
- $NP \rightarrow \text{Det } N \bullet$ [3]
- $PP \rightarrow P NP \bullet$ [2]
- $VP \rightarrow V PP \bullet$ [1]
- $S \rightarrow NP VP \bullet$ [0]
- $\gamma \rightarrow S \bullet$ [0]

1.3 Task 1 Conclusion

The presence of the completed state $\gamma \rightarrow S \bullet$ [0] in the final chart confirms successful parsing. The Earley parser efficiently handles lexical ambiguity (where words like *time*, *flies*, and *like* serve multiple parts of speech) by maintaining parallel states in a dynamic programming table.

2 Task 2: Probability of Each Parse Tree

2.1 Grammar Rule Probabilities

Rule	Prob	Rule	Prob
$S \rightarrow NP VP$	1.0	$N \rightarrow \text{time}$	0.4
$NP \rightarrow N N$	0.25	$N \rightarrow \text{flies}$	0.2
$NP \rightarrow D N$	0.4	$N \rightarrow \text{arrow}$	0.4
$NP \rightarrow N$	0.35	$D \rightarrow \text{an}$	1.0
$VP \rightarrow V NP$	0.6	$ADV \rightarrow \text{like}$	1.0
$VP \rightarrow V ADVP$	0.4	$V \rightarrow \text{flies}$	0.5
$ADVP \rightarrow ADV NP$	1.0	$V \rightarrow \text{like}$	0.5

2.2 Parse Tree 1: Most Natural Reading

Structure: [S [NP time] [VP flies [ADVP like [NP an arrow]]]]

Probability Calculation:

- $P(S \rightarrow NP VP) = 1.0$
- $P(NP \rightarrow N) = 0.35$
- $P(N \rightarrow \text{time}) = 0.4$
- $P(VP \rightarrow V ADVP) = 0.4$
- $P(V \rightarrow \text{flies}) = 0.5$
- $P(ADVP \rightarrow ADV NP) = 1.0$
- $P(ADV \rightarrow \text{like}) = 1.0$
- $P(NP \rightarrow D N) = 0.4$
- $P(D \rightarrow \text{an}) = 1.0$
- $P(N \rightarrow \text{arrow}) = 0.4$

$$P_1 = 1.0 \times 0.35 \times 0.4 \times 0.4 \times 0.5 \times 1.0 \times 1.0 \times 0.4 \times 1.0 \times 0.4 = \mathbf{0.00448}$$

2.3 Parse Tree 2: Alternative Reading

Structure: [S [NP time flies] [VP like [NP an arrow]]]

Probability Calculation:

- $P(S \rightarrow NP VP) = 1.0$
- $P(NP \rightarrow N N) = 0.25$
- $P(N \rightarrow \text{time}) = 0.4$
- $P(N \rightarrow \text{flies}) = 0.2$
- $P(VP \rightarrow V NP) = 0.6$

- $P(V \rightarrow \text{like}) = 0.5$
- $P(NP \rightarrow D N) = 0.4$
- $P(D \rightarrow \text{an}) = 1.0$
- $P(N \rightarrow \text{arrow}) = 0.4$

$$P_2 = 1.0 \times 0.25 \times 0.4 \times 0.2 \times 0.6 \times 0.5 \times 0.4 \times 1.0 \times 0.4 = \mathbf{0.00096}$$

2.4 Final Results and Conclusion

- **Parse Tree 1 Probability:** 0.00448
- **Parse Tree 2 Probability:** 0.00096

The analysis concludes that Parse Tree 1 (*time* as noun, *flies* as verb, *like* as preposition) is the more probable interpretation. The probabilistic grammar successfully reflects the intuitive human understanding of the sentence.

3 Task 3: Analysis of "the man shot the soldier with a gun"

3.1 Probabilistic Grammar (Figure 3)

The following Probabilistic Context-Free Grammar (PCFG) was used to parse the sentence:

Rule	Probability	Rule	Probability
$S \rightarrow NP VP$	1.0	$Det \rightarrow \text{the}$	1.0
$NP \rightarrow N$	0.35	$Det \rightarrow \text{a}$	1.0
$NP \rightarrow Det N$	0.4	$N \rightarrow \text{man}$	0.4
$NP \rightarrow N N$	0.25	$N \rightarrow \text{soldier}$	0.4
$VP \rightarrow V NP$	0.6	$N \rightarrow \text{gun}$	0.4
$VP \rightarrow V PP$	0.4	$V \rightarrow \text{shot}$	0.5
$PP \rightarrow P NP$	1.0	$P \rightarrow \text{with}$	1.0

3.2 Earley Chart Execution Highlights

The Earley parser processes the sentence through a series of charts, tracking the cumulative probability (p) of each state .

- **Chart[0] (Initialization):** The parser predicts the start symbol S and initial NP structures. The state $S \rightarrow \bullet NP VP$ [0] begins with $p = 1.0$.
- **Chart[2] (after "the man"):** The NP is completed via $NP \rightarrow Det N \bullet$ [0] with a cumulative probability of $p = 0.16$ ($1.0 \times 0.4 \times 0.4$) . This triggers the completion of $S \rightarrow NP \bullet VP$ [0], also at $p = 0.16$.
- **Chart[3] (after "shot"):** The verb is scanned. The state $VP \rightarrow V \bullet NP$ [2] is at $p = 0.30$ (0.6×0.5) relative to the VP start, while $VP \rightarrow V \bullet PP$ [2] is at $p = 0.20$ (0.4×0.5) .
- **Chart[5] (after "soldier"):** A partial parse is completed for the first five words. The state $S \rightarrow NP VP \bullet$ [0] reaches a probability of $p = 0.00768$.

Sentence: the man shot the soldier with a gun

Chart[0]

Y -> • S [0] (p=1.000000)
S -> • NP VP [0] (p=1.000000)
NP -> • N [0] (p=0.350000)
NP -> • Det N [0] (p=0.400000)
NP -> • N N [0] (p=0.250000)
N -> • man [0] (p=0.400000)
N -> • soldier [0] (p=0.400000)
N -> • gun [0] (p=0.400000)
Det -> • the [0] (p=1.000000)
Det -> • a [0] (p=1.000000)

Chart[1]

Det -> the • [0] (p=1.000000)
NP -> Det • N [0] (p=0.400000)
N -> • man [1] (p=0.400000)
N -> • soldier [1] (p=0.400000)
N -> • gun [1] (p=0.400000)

Chart[2]

N -> man • [1] (p=0.400000)
NP -> Det N • [0] (p=0.160000)
S -> NP • VP [0] (p=0.160000)
VP -> • V NP [2] (p=0.600000)
VP -> • V PP [2] (p=0.400000)
V -> • shot [2] (p=0.500000)

Chart[3]

V -> shot • [2] (p=0.500000)
VP -> V • NP [2] (p=0.300000)
VP -> V • PP [2] (p=0.200000)
NP -> • N [3] (p=0.350000)
NP -> • Det N [3] (p=0.400000)
NP -> • N N [3] (p=0.250000)
PP -> • P NP [3] (p=1.000000)
N -> • man [3] (p=0.400000)
N -> • soldier [3] (p=0.400000)
N -> • gun [3] (p=0.400000)
Det -> • the [3] (p=1.000000)
Det -> • a [3] (p=1.000000)
P -> • with [3] (p=1.000000)

Chart[4]

Det -> the • [3] (p=1.000000)
NP -> Det • N [3] (p=0.400000)
N -> • man [4] (p=0.400000)
N -> • soldier [4] (p=0.400000)
N -> • gun [4] (p=0.400000)

Chart[5]

N -> soldier • [4] (p=0.400000)
NP -> Det N • [3] (p=0.160000)
VP -> V NP • [2] (p=0.040000)
S -> NP VP • [0] (p=0.007600)
Y -> S • [0] (p=0.007600)

Chart[6]

Chart[7]

Chart[8]

Figure 1: This is the description of my chart.

3.3 Final Parsing Probability

The final state in Chart[5] indicates the probability for the recognized prefix/sentence segment:

$$P(\gamma \rightarrow S \bullet [0]) = 0.00768$$

3.4 Structural Ambiguity Note

The grammar includes rules for both $VP \rightarrow V NP$ and $VP \rightarrow V PP$. In the full sentence "the man shot the soldier with a gun," the chart shows the parser tracking multiple derivations to account for whether the prepositional phrase "with a gun" attaches to the Verb Phrase (the instrument of shooting) or the Noun Phrase (the soldier carrying the gun) .

4 Task 4: Code Architecture and Complexity

The implementation utilizes a Viterbi-integrated Earley parser to ensure both mathematical correctness in finding the most probable parse and computational efficiency.

4.1 Correctness: Tracking Derivations and Weights

To find the most probable parse tree, the parser integrates the Viterbi algorithm directly into the Earley states.

- **State Identity vs. Value:** An item's *identity* is defined by its structural position: the grammar rule, dot position, start index, and end index. Its *value* consists of its weight (negative log probability) and its backpointers to child states.
- **Tracking the Best Derivation:** New weights are calculated by summing the weights of antecedents. Because negative log probabilities are used, a lower weight indicates a higher actual probability.
- **Reacting to Collisions:** If a derivation is generated for an existing item, the parser compares weights. If the new derivation has a lower weight, a "Viterbi update" occurs; the existing weight and backpointers are overwritten, discarding the inferior derivation without pushing a duplicate state to the agenda.

4.2 Efficiency: Complexity Constraints

The dynamic programming architecture guarantees strict time and space bounds through the following mechanisms:

Metric	Complexity	Limiting Factor
Space Complexity	$O(n^2)$	Maximum unique states (span combinations).
Time Complexity	$O(n^3)$	The Completer operation searching for spans (i, j) and (j, k) .
Agenda Pushes	$O(1)$	Implementation of Hash Maps (Python dictionaries) for lookups.

Table 1: Computational Complexity Bounds

- **Space Complexity:** Space is bounded by the number of unique states. Since a state's identity is defined by start index i and end index j , there are at most $O(n^2)$ possible span combinations.
- **Time Complexity:** Dominated by the Completer. Pairing an active state (i, j) with a completed state (j, k) involves three indices ranging from 0 to n , resulting in $O(n^3)$ combinations.
- **Optimization via Hash Maps:** Utilizing a dictionary for the agenda allows $O(1)$ duplicate checks. Without this, linear scans would degrade the parser to $O(n^5)$ complexity, violating efficiency requirements.